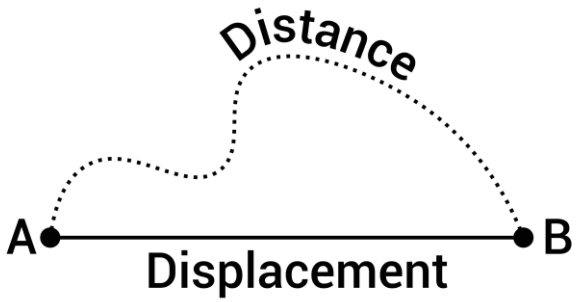
1. Law's of motion

1. The Fundamentals of Motion

Motion is a **relative concept**. An object is considered to be in motion if its position changes with respect to its surroundings; otherwise, it is at rest.

- **Distance:** The actual length of the path traveled by an object.
- **Displacement:** The minimum distance (straight line) between the starting and finishing points.

Note: Displacement can be zero if you return to your starting point, but the distance traveled will always be a positive value.



2. Speed and Velocity

These terms describe how fast an object moves, but they differ in directionality.

Feature	**Speed**	**Velocity**
Definition	Total distance divided by time	Displacement per unit time
Formula	$Speed = \frac{\text{Distance}}{\text{Time}}$	$Velocity = \frac{\text{Displacement}}{\text{Time}}$
Quantity Type	Scalar (Magnitude only)	Vector (Magnitude + Direction)
SI Unit	m/s	m/s

Factors affecting Velocity: Velocity changes if you change the speed, the direction of motion, or both.

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3. Acceleration

Acceleration is defined as the **rate of change of velocity**.

- **Formula:** $a = \frac{v - u}{t}$ (where v = final velocity, u = initial velocity, t = time).
- **Positive Acceleration:** Velocity increases over time.
- **Negative Acceleration (Deceleration/Retardation):** Velocity decreases; the force acts opposite to motion.
- **Zero Acceleration:** Velocity remains constant.

4. Newton's Equations of Motion

Newton provided three fundamental equations for objects moving in a straight line with uniform acceleration:

1. **Velocity-Time Relation:** $v = u + at$
2. **Displacement-Time Relation:** $s = ut + \frac{1}{2}at^2$
3. **Displacement-Velocity Relation:** $v^2 = u^2 + 2as$

5. Uniform Circular Motion

When an object moves at a constant speed along a circular path, it is in **Uniform Circular Motion**.

- **Changing Velocity:** Even if speed is constant, velocity is always changing because the **direction** changes at every point.
- **Direction:** The velocity at any point is always **tangent** to the circular path.
- **Calculation:** $v = \frac{2\pi r}{t}$

6. Newton's Three Laws of Motion

First Law: Law of Inertia

An object remains at rest or in uniform motion unless acted upon by an external unbalanced force. **Inertia** is the inherent resistance of an object to change its state.

Second Law: Force and Momentum

The rate of change of momentum is proportional to the applied force.

- **Momentum (P):** $P = mv$ (Mass × Velocity).
- **Force Formula:** $F = m \times a$
- **Units:** SI unit is Newton (N); CGS unit is dyne ($1 N = 10^5 dynes$).

Third Law: Action and Reaction

For every action, there is an equal and opposite reaction acting simultaneously.

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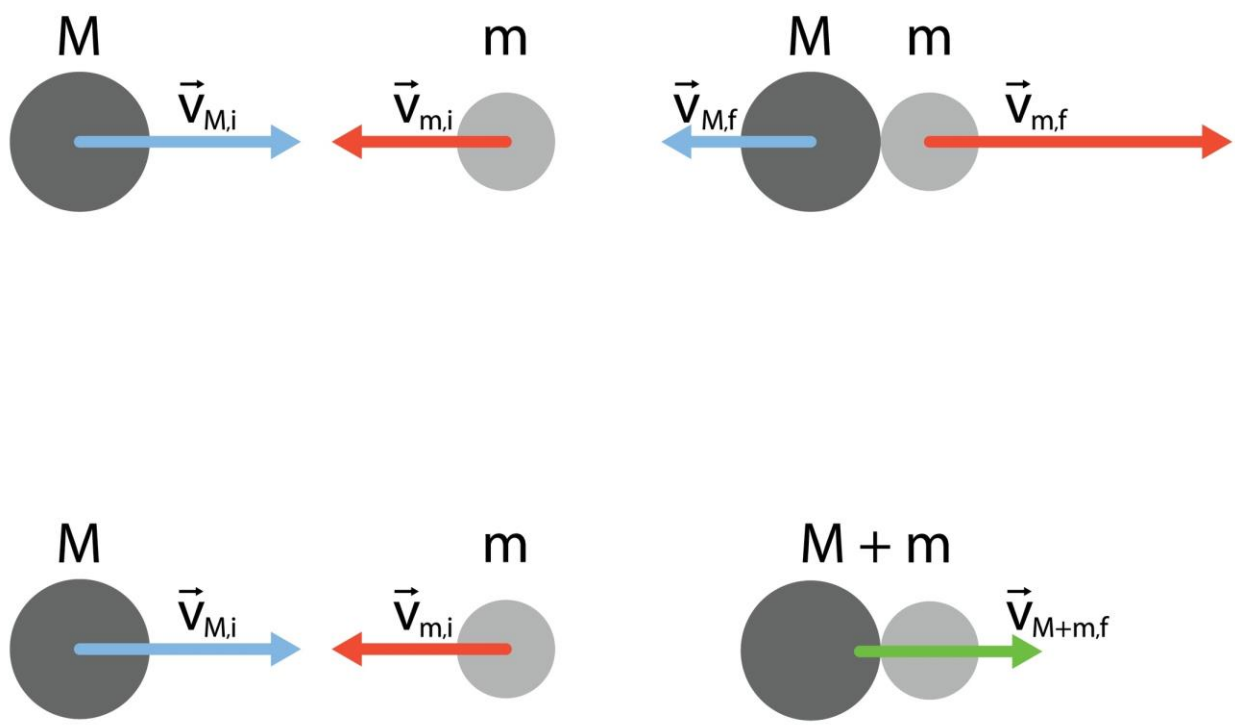
- These forces act on **different objects**, so they do not cancel each other out.

7. Law of Conservation of Momentum

If no external force acts on two interacting objects, their total momentum remains constant.

- **Collision Formula:** $m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$
- **Recoil Example:** When a gun fires a bullet, the gun moves backward (recoils). The total momentum before and after the shot remains zero because the forward momentum of the bullet is balanced by the backward momentum of the gun.



1. Law's of motion

- Laws of Motion
  - Basic Concepts of Motion
    - Motion
      - Relative Concept
      - Change in Position
    - Distance and Displacement
      - Distance: Actual path length
      - Displacement: Minimum distance
    - Speed and Velocity
      - Speed: Total distance / Time
      - Velocity: Displacement / Time
      - Scalars and Vectors
  - Types of Motion
    - Linear Motion
      - Uniform: Equal distance in equal time
      - Non-uniform: Unequal distance
    - Uniform Circular Motion
      - Constant Speed
      - Changing Direction
      - Velocity Tangential
  - Acceleration
    - Rate of change of velocity
    - Formula: a = (v - u) / t
    - Types
      - Positive Acceleration
      - Negative (Deceleration)
      - Zero Acceleration
  - Newton's Laws of Motion
    - First Law (Inertia)
      - Balanced and Unbalanced Forces
      - State of Rest or Uniform Motion
    - Second Law (Momentum)
      - Momentum (P) = m x v
      - Force (F) = m x a
      - Rate of change of momentum
    - Third Law (Action-Reaction)
      - Equal and Opposite Reaction
      - Acts on different objects
  - Equations of Motion
    - v = u + at
    - s = ut + 1/2 at^2
    - v^2 = u^2 + 2as
  - Conservation of Momentum
    - Total initial = Total final
    - External force is zero
    - Recoil of gun